

Practice Tests 2

Course Name Database Principle Examination Time 120minutes

Student Number _____ Student Name _____

1. Answer the following questions: (4+4+4+4+4)
 - (1) In database design, is the higher the NF(Normal Form) that the data schema follows, the better?
 - (2) Explain the ACID properties of “Transaction”?
 - (3) List those sub-constraints included in Static constraints. Which sub-constrain contains Domain constraints? And explain what’s Domain constraints.
 - (4) List three recovery strategies in DBMS, and describe the necessity of the commit rule and the log ahead rule for updating transactions.
 - (5) Does dense index always improve the efficiency of queries on the indexed attribute? Why?
2. Try to answer the following questions about concurrency in database systems.
 - (1) What is concurrency?(1)
 - (2) Why concurrency?(1)
 - (3) What problems will concurrency bring?(2)
 - (4) What kind of concurrent execution is correct?Why?(3)
 - (5) How to avoid problems caused by concurrency?(3)
3. What is the basic idea of the locking protocol and how does it ensure that concurrent transactions are executed correctly? What are the problems that must be solved after the adoption of the locking protocol? (5+5)
4. Consider a relational table:
Professor(
 professor_name, professor_id,
 professor_office_id, student_id,
 student_name, student_office_id,
 student_designated_refrigerator_id, refrigerator_owner_id,
 refrigerator_id, refrigerator_size, secretary_name,
 secretary_id, secretary_office)
Suppose the data has the following properties:
 - A. Professors and secretaries have individual offices, students share offices.
 - B. Students can work for multiple professors.
 - C. Refrigerators are owned by one professor.
 - D. Professors can own multiple refrigerators.
 - E. Students can only use one refrigerator.
 - F. The refrigerator the student uses must be owned by one of the professors they

work for.

G. Secretaries can work for multiple professors.

H. Professors only have a single secretary (

- (1) Put this table into 3rd normal form by writing out the decomposed tables; designate keys in your tables by underlining them. Designate foreign keys by drawing an arrow from a foreign key to the primary key it refers to. Note that some of the properties listed above may not be enforced (i.e., guaranteed to be true) by a 3NF decomposition.(5)
- (2) Which of the eight properties (A–H) of the data are enforced (i.e., guaranteed to be true) by the 3NF decomposition and primary and foreign keys you gave above.(5)
- (3) Please draw the E-R diagram of the relationship in this question.(10)

5. Hotel Management System introduction:

One room have one or several beds, and each room has information of room ID (e.g., 201), price and bed number.

Each customer has information of ID, name and gender.

Each reserve record has information of the ID of customer, the ID of room and reserve time.

Each room can be reserved by only one customer at the same time.

Table:

Room(RID, Price, Number);

Customer(CID, Name, Gender);

Reserve(RID, CID, Rdate)

- (1) For the relationships in the this question, we have the following query(10):

```
SELECT Name
```

```
FROM Room, Customer, Reserve
```

```
WHERE Room.RID = Reserve.RID AND
```

```
Customer.CID = Reserve.CID AND
```

```
Room.Price > 500 AND
```

```
Customer.Gender = 'female' AND
```

```
Reserve.Rdate between [2019.10.01, 2019.10.07]
```

Please optimize the query and draw the query trees before and after optimization.

- (2) For the Room table in this question, apply Assertion to ensure that the number of rooms ≥ 0 . (5)

6. For the three relational schemas in the above application scenario, write SQL statements that express the following query requirements (must be expressed in a single SQL statement, and WITH is only accepted in question):

- (1) Use join query to find the name of the guest who booked room 210;(5)
- (2) Query the name of the guest who booked all rooms;(5)
- (3) Query the room number and guest name booked by only one person in January 2016;(10)

- (4) For guests who have booked a room more than 60 times, query the maximum number of room bookings per guest in a single month and the year and month corresponding to the maximum number.(10)

Note: The functions for obtaining the year and month of Datetime data in SQL are getYear(...) and getMonth(...) respectively.

7. The compatibility matrix of (S, X) locks is shown in Figure (a). Suppose that we change the compatibility matrix of (S, X) locks to the one as shown in Figure (b). Now we have n transactions $\{T_1, T_2, \dots, T_n\}$ that require concurrent access in a period of time. Assume that these n transactions are read operations. Will the system work properly? If yes, explain the reason; If not, what happens? (10)

	Locks already owned by other transactions		
		S	X
	S	Y	N
	X	N	N

(a) Compatibility Matrix A

	Locks already owned by other transactions		
		S	X
	S	Y	Y
	X	N	N

(b) Compatibility Matrix B

8. Please answer the following questions: (5+5+10=20)

- (1) Please explain what is Trigger and list the components of Trigger.
- (2) Design a trigger for the SelectCourse table (in the fourth question) to ensure that $0 \leq \text{Grade} \leq 100$ and Sno exists in Student table and Cno exists in Course table when a tuple is inserted.
- (3) Try to define a trigger on the accommodation relationship table in question 2 to achieve the following function: For each Insert statement, if the inserted record does not have a check-out date, the operation will be rolled back.